## Type MS · Arc length, sector area and perimeter

Answer 1 of 11 · Question 1 part b · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | |
|---|---|---|
| **1(a)** | $A = \frac{1.2}{2} \times 7^2 = 29\frac{2}{5}$ | M1A1 [2] |
| **(b)** | $P = 7+7+7\times1.2 = 22\frac{2}{5}$ (cm) | M1A1 [2] |
| | **Total** | **4 marks** |

| Part | Mark | Notes |
|---|---|---|
| **(a)** | M1 | For using the correct formula for the area of a sector with correct substitution of the given values. $A = \frac{1.2}{2} \times 7^2 = ...$ |
| | A1 [2] | For $A = 29\frac{2}{5}$ o.e. ($cm^2$) |
| **(b)** | M1 | For a complete method to find the perimeter of the sector with correct substitution of the given values $P = 7+7+7\times1.2 = ...$ |
| | A1 [2] | $P = 22\frac{2}{5}$ o.e. (cm) |

## Type MS · Arc length, sector area and perimeter

### Answer 1 of 11 · Question 1 part b · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question | Scheme | Mark |
|---|---|---|

Type MS · Arc length, sector area and perimeter

Answer 2 of 11 · Question 1 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

**Use of a formula:**

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**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

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Type MS · Arc length, sector area and perimeter

Answer 2 of 11 · Question 1 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

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Type MS · Arc length, sector area and perimeter

Answer 2 of 11 · Question 1 part a · 1 mark

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

**June 2019**
**4PM1 Further Pure Mathematics Paper 1**

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $l = r\theta \Rightarrow r = \frac{12}{1.5} = 8$ | B1 [1] |
| (b) | $A = \frac{1.5}{2} \times 8^2 = 48\ (\text{cm}^2)$ | M1A1 [2] |
| | **ALT 1** $A = \frac{l^2}{2\theta} = \frac{12^2}{2 \times 1.5} = 48\ (\text{cm}^2)$ | {M1A1} [2] |
| | **ALT 2** $A = \frac{1}{2}rl = \frac{1}{2} \times 8 \times 12 = 48\ (\text{cm}^2)$ | {M1A1} [2] |
| | | Total 3 marks |

Type MS · Arc length, sector area and perimeter

Answer 3 of 11 · Question 1 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

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| | **ALT 1** $A = \frac{l^2}{2\theta} = \frac{12^2}{2 \times 1.5} = 48\ (\text{cm}^2)$ | {M1A1} [2] |
| | **ALT 2** $A = \frac{1}{2}rl = \frac{1}{2} \times 8 \times 12 = 48\ (\text{cm}^2)$ | {M1A1} [2] |
| | | Total 3 marks |

Type MS · Arc length, sector area and perimeter

Answer 4 of 11 · Question 1 part a · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| | | |
|---|---|---|
| **1(a)** | $A = \frac{1.2}{2} \times 7^2 = 29\frac{2}{5}$ | M1A1 [2] |
| **(b)** | $P = 7+7+7\times1.2 = 22\frac{2}{5}$ (cm) | M1A1 [2] |
| | **Total** | **4 marks** |

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|---|---|---|
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| | A1 [2] | For $A = 29\frac{2}{5}$ o.e. ($cm^2$) |
| **(b)** | M1 | For a complete method to find the perimeter of the sector with correct substitution of the given values $P = 7+7+7\times1.2 = ...$ |
| | A1 [2] | $P = 22\frac{2}{5}$ o.e. (cm) |

## Type MS · Arc length, sector area and perimeter

### Answer 4 of 11 · Question 1 part a · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question | Scheme | Mark |
|---|---|---|

## Type MS · Arc length, sector area and perimeter

### Answer 5 of 11 · Question 3 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

| | | |
|---|---|---|
| 3 (a) | $(\frac{1}{2} \times 4^2 \times \theta = 2\pi)$<br><br>$\frac{1}{2} \times$ " $16$" $\times \theta = 2\pi \Rightarrow \theta = ...$<br><br>$\theta = \frac{2\pi}{8} = \frac{\pi}{4}$ oe | M1<br><br>A1<br><br>[2] |
| (b) | (arc $SR$ = ) $4 \times$ " $\frac{\pi}{4}$ " $(= \pi)$<br><br>$(4 + 4 + \pi) = 11.1$ (cm) Accept awrt 11.1 (cm) | M1<br><br>A1ft<br>[2] |
| (c) | $\frac{1}{2} \times (1) \times 4 \times \sin$ " $\frac{\pi}{4}$ " $(= \sqrt{2})$<br><br>$2\pi - \sqrt{2}$ oe | M1<br><br>A1<br>[2] |
| | | **Total 6 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use of the correct formula for the area of sector, with correct substitution of 4 and $2\pi$, an attempt to square $r$ and to rearrange to find a value for $\theta$.<br>The rearrangement has to be correct for this mark. |
| | A1 | Accept any valid exact equivalent.<br>Do not accept answers in degrees or answers not given exactly for A1.<br>However, $45^o$ or 0.785(3981…) stated with no method - award implied M1. |
| (b) | M1 | Use of correct formula for length of an arc with the correct radius and "their $\theta$" (allow working in degrees for this mark) from part (a).<br>Do **not** allow $\theta = \frac{\pi}{2}$ or $90^o$ |
| | A1ft | Allow **ft** of their angle from part (a), if complete method shown and if their answer is correct to 3sf.<br>**Note this is ft mark. Check working carefully.** |
| (c) | M1 | Use of correct formula for area of a triangle with correct sides substituted and "their $\theta$" (allow working in degrees) from part (a).<br>Do **not** allow $\theta = \frac{\pi}{2}$ or $90^o$ |
| | A1 | For the correct answer shown. |

## Type MS · Arc length, sector area and perimeter

### Answer 5 of 11 · Question 3 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Arc length, sector area and perimeter

### Answer 6 of 11 · Question 3 part b · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | |
|---|---|---|
| 3 (a) | $(\frac{1}{2} \times 4^2 \times \theta = 2\pi)$<br><br>$\frac{1}{2} \times$ " $16$" $\times \theta = 2\pi \Rightarrow \theta = ...$<br><br>$\theta = \frac{2\pi}{8} = \frac{\pi}{4}$ oe | M1<br><br>A1<br><br>[2] |
| (b) | (arc $SR$ = ) $4 \times$ " $\frac{\pi}{4}$ " $(= \pi)$<br><br>$(4 + 4 + \pi) = 11.1$ (cm) Accept awrt 11.1 (cm) | M1<br><br>A1ft<br>[2] |
| (c) | $\frac{1}{2} \times (1) \times 4 \times \sin$ " $\frac{\pi}{4}$ " $(= \sqrt{2})$<br><br>$2\pi - \sqrt{2}$ oe | M1<br><br>A1<br>[2] |
| | | **Total 6 marks** |

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | M1 | Use of the correct formula for the area of sector, with correct substitution of 4 and $2\pi$, an attempt to square $r$ and to rearrange to find a value for $\theta$.<br>The rearrangement has to be correct for this mark. |
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| | A1 | For the correct answer shown. |

## Type MS · Arc length, sector area and perimeter

### Answer 6 of 11 · Question 3 part b · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Arc length, sector area and perimeter

Answer 7 of 11 · Question 1 part a · 1 mark

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

1. Factorisation:

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Type MS · Arc length, sector area and perimeter

Answer 7 of 11 · Question 1 part a · 1 mark

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

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Type MS · Arc length, sector area and perimeter

Answer 7 of 11 · Question 1 part a · 1 mark

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

**MARK SCHEME**

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $l = r\theta \Rightarrow l = 13 \times 2 = 26$ (cm) | B1 [1] |
| (b) | $A = \frac{\theta}{2} r^2 = \frac{2}{2} \times 13^2 = 169$ (cm$^2$) | M1A1 [2] |
| | | **Total 3 marks** |

Type MS · Arc length, sector area and perimeter

Answer 8 of 11 · Question 1 part b · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

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Answer 8 of 11 · Question 1 part b · 2 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

**MARK SCHEME**

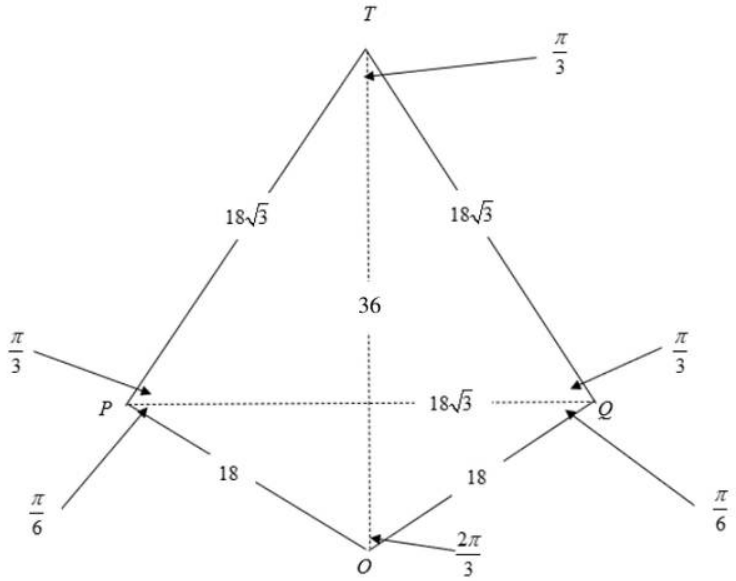
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## Type MS · Arc length, sector area and perimeter

### Answer 9 of 11 · Question 3 part a · 2 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| 3 (a) | $PQ = 18 \times \frac{2}{3}\pi = 12\pi$ | M1 A1 (2) |
| (b)(i) | $\alpha = \frac{1}{3}\pi$ | B1 |
| (b)(ii) | $PT = 18\tan\frac{\pi}{3} = 18\sqrt{3}$ | M1 A1 |
| | Area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3}$ | M1 |
| | Area of Sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3}$ | M1 |
| | Shaded Area $= 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3} - \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = 222 \text{ cm}^2$ | M1 A1 (7) |
| | | **Total 9 marks** |




Area of triangle $OPQ = 81\sqrt{3}$ or 140.29... $\text{cm}^2$

Area of triangle $PQT = 243\sqrt{3}$ or 420.88... $\text{cm}^2$

Area of quadrilateral $OTPQ = 324\sqrt{3}$ or 561.18... $\text{cm}^2$

| Part | Mark | Guidance |
|---|---|---|
| **(a)** | **M1** | Uses the **correct** formula for the length of arc to give $PQ = 18 \times \frac{2}{3}\pi = ...$ |
| | **A1** | For $PQ = 12\pi$ |

Type MS · Arc length, sector area and perimeter

Answer 9 of 11 · Question 3 part a · 2 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

| | | |
|---|---|---|
| | **ALT – works in degrees** (but the angle must be correct at $120^o$) | |
| | **M1** | Uses the **correct** formula for length of arc to give<br>$PQ = \frac{120}{360} \times 2\pi \times 18 = ...$ |
| | **A1** | For $PQ = 12\pi$ |
| **(b)** | **(i) B1** | For stating $\alpha = \frac{\pi}{3}$ (Please check the diagram as it may written on there) |
| | **(ii) Method 1** - Allow use of degrees throughout provided the angles are correct.<br>$\left(\angle POQ = 120^o,\ \angle PTQ = 60^o\right)$ | |
| | **Finds length of *PT*** | |
| | **M1** | For finding length $PT$ : e.g.,<br>$\tan\left(\frac{\pi}{3}\right) = \frac{PT}{18} \Rightarrow PT = 18\tan\left(\frac{\pi}{3}\right) = ...$<br>The given values must be used correctly |
| | **A1** | For $PT = 18\sqrt{3}$ |
| | **M1** | For the area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times '18\sqrt{3}' = \left(561.18...\right)$<br>Their $18\sqrt{3}$ must come from an attempt at using trigonometry. |
| | **Method 2** | |
| | **Finds lengths *PQ* and *TO*** | |
| | **M1** | For finding the lengths $PQ$ **and** $TO$ using any acceptable correct trigonometry.<br>e.g., $PQ = \sqrt{18^2 + 18^2 - 2 \times 18 \times 18\cos\left(\frac{2\pi}{3}\right)} = ...$ and<br>$TO = \frac{\cos\left(\frac{\pi}{3}\right)}{18} = ...$<br>The given values must be used correctly |
| | **A1** | For both correct lengths: $PQ = 18\sqrt{3}$ and $TO = 36$ |
| | **M1** | For the area of $OPTQ = \frac{1}{2} \times '18\sqrt{3}' \times '36' = \left(561.18...\text{ or } 324\sqrt{3}\right)$<br>Their $18\sqrt{3}$ and $36$ must come from an attempt at using trigonometry |
| | | |
| | **M1** | For the area of sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = \left(339.29...\text{ or } 108\pi\right)$<br>or $\frac{120^o}{360^o} \times \pi \times 18^2 = \left(339.29...\right)$ |
| | **M1** | For area of $OPTQ$ – area of Sector $OPQ$<br>$561.18 - 339.29 = 221.887\ldots$ |
| | **A1** | For $222\ \text{cm}^2$ (must be 3sf) (Units are not required) |

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| Question number | Scheme | Marks |
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Type MS · Arc length, sector area and perimeter

Answer 10 of 11 · Question 1 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

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Type MS · Arc length, sector area and perimeter

Answer 10 of 11 · Question 1 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

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### Answer 10 of 11 · Question 1 part a · 2 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $\frac{1}{2}2^2\theta=\frac{\pi}{2}\Rightarrow\theta=$<br>$\theta=\frac{\pi}{4}$ | M1<br>A1<br>[2] |
| (b) | $2\times"\frac{\pi}{4}"\left(=\frac{\pi}{2}\right)$<br>$"\frac{\pi}{2}"+2+2$ or $"\frac{\pi}{2}"+2\times2$ or $"\frac{\pi}{2}"+4$<br>$4+\frac{\pi}{2}$ oe | M1<br>M1<br>A1<br>[3] |
| | | **Total 5 marks** |

Type MS · Arc length, sector area and perimeter

Answer 11 of 11 · Question 1 part b · 3 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

**Method marks for differentiation and integration:**

1. Differentiation

Power of at least one term decreased by 1. $(x^n\rightarrow x^{n-1})$

2. Integration:

Power of at least one term increased by 1. $(x^n\rightarrow x^{n+1})$

**Use of a formula:**

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

**or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

**Answers without working:**

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Arc length, sector area and perimeter

Answer 11 of 11 · Question 1 part b · 3 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

**Rounding answers (where accuracy is specified in the question)**

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

## Type MS · Arc length, sector area and perimeter

### Answer 11 of 11 · Question 1 part b · 3 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| 1 (a) | $\frac{1}{2}2^2\theta=\frac{\pi}{2}\Rightarrow\theta=$<br>$\theta=\frac{\pi}{4}$ | M1<br>A1<br>[2] |
| (b) | $2\times"\frac{\pi}{4}"\left(=\frac{\pi}{2}\right)$<br>$"\frac{\pi}{2}"+2+2$ or $"\frac{\pi}{2}"+2\times2$ or $"\frac{\pi}{2}"+4$<br>$4+\frac{\pi}{2}$ oe | M1<br>M1<br>A1<br>[3] |
| | | **Total 5 marks** |